

RNA target sequence

Linear probing

stem vs. loop classification (BA)

TWO-STAGE GATE

FAIL

$R < p_{95}$

**Gate B: Composition null**

Nucleotide-stratified + dinucleotide-stratified  
1,000 permutations,  $p_{95}$  threshold

Sensitivity ratio within  
composition null  
(14–16 of 16 targets)

$R > p_{95}$

PASS

FAIL

$\rho_S < 0.10$

**Gate C: Attention–contact**

Best-head  $\rho_S$  vs. ViennaRNA contact map  
Threshold:  $\rho_S \geq 0.10$

$\rho_S \geq 0.10$

PASS

“Degeneracy trap”  
(H-E3: 2 instances)

**Untrained-weight control**

5-seed random init, same architecture

Trained – untrained  $\Delta$  with 95% CI

NTv2:  $\Delta = -0.002$ , CI straddles zero  
→ fully architectural artifact

DNABERT-2:  $\Delta = +0.029$ , CI excludes zero  
→ small learned increment (12% of base)